MIL-R-25534A(USAF)
6 April 1959
Superseding
MIL-E-25534(USAF)
14 March 1956

MILITARY SPECIFICATION

ROCKET MOTORS, AERONAUTICAL, QUALIFICATION TEST FOR

1. SCOPE

1.1 This specification establishes the qualification test requirements for rocket motors for aeronautical vehicles.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification.

SPECIFICATIONS

Military

MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation For Delivery
MIL-P-8564	Pneumatic Components, Aeronautical, General Specification For
MIL-H-8775	Hydraulic System Components, Aircraft, General Specification For
MIL-T-9107	Test Reports, Preparation Of
MIL-H-25475	Hydraulic Systems: Design, Installation, and Tests Of Missile (General Specification For)
MIL-R-25532	Rocket Motors, Aeronautical, General Specification For

STANDARDS

Military

MIL-STD-414	Sampling Procedures and Tables For Inspection By Variables For Percent Defective

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 1336
FSC 1340

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3. REQUIREMENTS

3.1 REPORTS, ROCKET MOTOR AND COMPONENTS.- Reports shall be prepared as follows:

3.1.1 GENERAL

3.1.1.1 DIMENSIONAL UNITS.- All dimensions shall be expressed in the English gravitational system of units.

3.1.1.2 TEST DATA CORRECTIONS

3.1.1.2.1 TEMPERATURE.- Performance characteristics shall not be corrected for free air temperature.

3.1.1.2.2 PRESSURE.- An approximate correction for barometric pressure altitude shall be applied to thrust and impulse values. This correction shall be made as follows:

$$F_{sl} = F_{alt} - A_e(P_{sl} - P_{alt}) \qquad I_{sl} = I_{alt} \times \frac{F_{sl}}{F_{alt}}$$

3.1.1.3 DATA SUMMARY SHEET.- A data summary sheet shall be completed for each rocket motor firing. This sheet shall be prepared in accordance with figure 1. Where the length of traces makes their inclusion difficult, only the initial and final portions, and any other portions of significant interest shall be included.

3.1.1.3.1 LIMITS.- Performance limits shall be superimposed on all curves. These limits shall be the same as those specified in the model specification.

3.1.1.4 STATISTICAL SUMMARY SHEETS.- A statistical summary sheet based on the data resulting from the firings of test groups I thru VIII shall be prepared for each parameter in the sea level performance rating table of the model specification. These sheets shall be prepared in accordance with figure 2. The acceptability of each parameter shall be determined in accordance with the procedures of MIL-STD-414 as follows:

a. The data from test groups I, II, III, VII, and VIII shall be reduced, using the procedure of example B-2 of MIL-STD-414 for single specification limits and the procedure of example B-3 for double specification limits. (Table B-5 of MIL-STD-414 shall be used to estimate the percent defective.) Each parameter shall be deemed acceptable at a given temperature if the estimated percent defective is less than or equal to the maximum allowable percent defective, M, as shown in table I for the sample size involved.

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FIGURE 1

Data Summary Sheet

Test conducted by (name of contractor) Date_____

Contract No._____ Purpose_____

I ROCKET MOTOR INFORMATION

Rocket Motor Model Designation _____
Rocket Motor Lot Number _____ Rocket Motor Sublot Number _____
Rocket Motor Serial No. _____ Case Serial No. _____
Nozzle Serial No. _____
Propellant Type or Designation _____
Propellant Lot No. _____ Propellant Batch No.(s) _____
Propellant Grain Serial No. _____

II IGNITER INFORMATION

Type _____
Lot No. _____ Sublot No. _____ Igniter Serial No. _____
Squib Resistance Before Conditioning _____ After Conditioning _____

III PHYSICAL DATA

Rocket Motor Weight - before firing _____ Center of Gravity Location
- after firing _____ - before firing _____
- after firing _____

Nozzle Expansion Ratio _____
Nozzle Throat Area - before firing _____
- after firing _____

IV ENVIRONMENTAL AND PRETEST TREATMENT

Conditioned Temperature _____
Temperature cycled from _____ to _____ for _____ times.
Vibrated for _____ hours at _____ cps parallel, perpendicular (specify)
to the longitudinal axis. (Repeat this statement as necessary.)
Rough road tested _____
Uncrated - dropped from 4 ft. impact on _____.
Crated - edgewise dropped from _____ inches.
Crated - cornerwise dropped from _____ inches.
Crated - pendulum, incline (specify) from _____ inches.
Altitude cycled from _____ in Hg to _____ in Hg to _____ in Hg for
_____ times.

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FIGURE 1
(continued)

IV ENVIRONMENTAL AND PRETEST TREATMENT (continued)

Humidity _____
Rain _____
Salt Spray _____
Sand and Dust _____
Accelerated Aging Hot _____ Cold _____
Temperature Gradient - low to high _____

V ACTUAL TEST CONDITIONS

Rocket Motor Conditioning Temperature _____
Time removed from conditioning _____ time fired _____
Observed Barometric Pressure _____
Temperature of Barometer _____
Corrected Barometric Pressure _____
Ambient Air Temperature at firing _____

VI TEARDOWN INSPECTION RESULTS

(Include results of teardown inspection.)

VII REDUCED DATA

Action time total impulse _____
Burn time total impulse _____
Deliverable total impulse _____
Specific impulse _____
Overall specific impulse _____
Maximum thrust _____
Maximum ignition thrust _____
Average thrust _____
Maximum chamber pressure _____
Maximum ignition pressure _____
Average chamber pressure _____
Action time _____
Burn time _____
Ignition delay time _____
Ignition rise time _____
Burn rate _____
Igniter current used _____
Thrust Angle Control Device.- Curves showing the variation of actuating torque, jet deflection, loss of action time total impulse, and normal force versus control deflection.

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FIGURE 1
(continued)

VII REDUCED DATA (continued)

Thrust Magnitude Control Device.- Curves plotted showing the variation in thrust change versus response time and actuating torque versus thrust.
Thrust Termination Device.- Curve plotted showing thrust versus elapsed time after termination signal.
Thrust Reversal Device.- Curve plotted showing thrust versus elapsed time after reversal signal.

VIII PERFORMANCE TRACES

Thrust-time trace with point of firing switch actuation shown
Pressure-time trace with point of firing switch actuation shown
Hi-speed ignition thrust-time trace with point of firing switch actuation shown
Hi-speed ignition pressure-time trace with point of firing switch actuation shown

IX THRUST CONTROL DEVICES

Device Model Designation _____
Device Lot Number _____ Device Sublot Number _____
Device Serial Number _____
Squib Resistance _____
Required Component Testing Completed (specify yes or no)

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FIGURE 2

Statistical Summary Sheet

Parameter - (Specify parameter)

Specification Limit			
Maximum	_____	_____	_____
Minimum	_____	_____	_____

Group I		Group II		Group III	
#1	_____	#1	_____	#1	_____
#2	_____	#2	_____	#2	_____
#3	_____	#3	_____	#3	_____
#4	_____			#4	_____
$\bar{X}_4$	_____	$\bar{X}_3$	_____	$\bar{X}_4$	_____
S_4	_____	S_3	_____	S_4	_____
p	_____	p	_____	p	_____
M_4	1.53	M_3	7.59	M_4	1.53

Accepted (Specify yes or no)(Specify yes or no)(Specify yes or no)

Group IV		Group V		Group VI	
#5	_____	#4	_____	#5	_____
#6	_____	#5	_____	#6	_____
#7	_____	#6	_____	#7	_____
#8	_____	#7	_____	#8	_____
#9	_____			#9	_____
#10	_____			#10	_____
$\bar{X}_{10}$	_____	$\bar{X}_7$	_____	$\bar{X}_{10}$	_____
S_{10}	_____	S_7	_____	S_{10}	_____
p	_____	p	_____	p	_____
M_{10}	0.349	M_7	0.422	M_{10}	0.349

Accepted (Specify yes or no)(Specify yes or no)(Specify yes or no)

NOTE: First half of rocket motors listed in test groups IV, V, and VI were subjected to humidity and rain; second half were subjected to salt spray and sand and dust.

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FIGURE 2
(continued)

	Group VIIA	Group VIIB
#1	_____	_____
#2	_____	_____
#3	_____	_____
$\bar{X}_3$	_____	_____
S_3	_____	_____
p	_____	_____
M_3	7.59	7.59
Accepted	(Specify yes or no)	(Specify yes or no)

	Group VIII
#1	_____
#2	_____
#3	_____
$\bar{X}_3$	_____
S_3	_____
p	_____
M_3	7.59
Accepted	(Specify yes or No)

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b. The data for test groups IV, V, and VI shall be grouped with the data for test groups I, II, and III, respectively, and reduced as specified in a. Each parameter shall be deemed acceptable with regard to sequential treatment at a given temperature if the estimated percent defective is less than or equal to the maximum allowable percent defective, M, as shown in table I for the sample size involved.

TABLE I

Maximum Allowable Percent Defective, M

Sample Size	M
3	7.59
4	1.53
7	0.422
10	0.349

3.1.2 PRELIMINARY REPORT.- If requested by the procuring activity, a brief preliminary report certified by a representative of the procuring activity as to proper conduct of the tests and factual accuracy shall be submitted immediately following each separate set of component tests and completion of tests on any group of rocket motors. At the discretion of the procuring activity, these reports may constitute the basis for approval of the qualification test. The preliminary report shall contain, but not necessarily be limited to, the following:

3.1.2.1 TITLE PAGE.- (Title of report, report number, report source, date, name(s) of author(s), and contract number)

3.1.2.2 TABLE OF CONTENTS

3.1.2.3 ABSTRACT.- (A brief resumé of the report stating the purpose, conclusions, and recommendations. Conclusions and recommendations not found in the report proper shall not be included in the abstract.)

3.1.2.4 BODY.- (A complete discussion, including results, dates, incidents or failures, and marginal conditions encountered)

3.1.2.5 CONCLUSIONS AND RECOMMENDATIONS.- (Include such discussion as required for justification.)

3.1.2.6 APPENDIX I DATA SUMMARY SHEETS.- (Include a data summary sheet for each test firing.)

3.1.2.7 APPENDIX II COMPONENT DATA.- (Include data on components tested during rocket motor firings.)

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3.1.2.8 APPENDIX III ACCEPTANCE DATA.- (Include acceptance data on all rocket motors and components.)

3.1.2.9 APPENDIX IV REJECTION DATA.- (Include data on any rocket motors or components rejected during acceptance testing.)

3.1.3 FINAL REPORT.- Following completion of all rocket motor tests required herein, a final report shall be submitted which will constitute a record of all information pertaining to the rocket motor tests. This report, in conjunction with the component test reports, will normally be used as a basis for approval of the qualification test. The final report shall contain, but not necessarily be limited to, the following:

3.1.3.1 TITLE PAGE.- (Title of report, report number, report source, date, name(s) of author(s), and contract number)

3.1.3.2 ABSTRACT.- (A brief resume of the report, stating the purpose, conclusions, and recommendations. Conclusions and recommendations not found in the report proper shall not be included in the abstract.)

3.1.3.3 TABLE OF CONTENTS

3.1.3.4 OBJECT

3.1.3.5 CONCLUSIONS AND RECOMMENDATIONS.- (Include such discussion as required for justification.)

3.1.3.6 DESCRIPTION.- (A brief general description of the rocket motor and components and a detailed description of all features that differ from the previous model, if applicable.)

3.1.3.7 METHOD.- (Reference approved qualification test method that was submitted with the model specification and detail any differences between the approved method and the actual method.)

3.1.3.8 RECORD OF TEST.- (Chronological history of all events connected with the testing)

3.1.3.9 ANALYSIS OF RESULTS.- (A complete discussion of all phases of the testing, including a comparison of demonstrated performance with that contained in the model specification. Statistical summary sheets for each parameter in the sea level performance rating table of the model specification shall be included.)

3.1.3.10 APPENDIX I DATA SUMMARY SHEETS.- (Include a data summary sheet for each test firing.)

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3.1.3.11 APPENDIX II COMPONENT DATA.- (Include data on components tested during rocket motor firings.)

3.1.3.12 APPENDIX III ACCEPTANCE DATA.- (Include acceptance data on all rocket motors and components.)

3.1.3.13 APPENDIX IV REJECTION DATA.- (Include data on any rocket motors or components rejected during acceptance testing.)

3.1.4 COMPONENT TEST REPORTS.- Separate component test reports covering all component tests conducted on each type of component (i.e., thrust angle control, electrical system, case, etc.) shall be submitted. These reports shall be prepared in accordance with MIL-T-9107.

3.1.5 NUMBER AND DISTRIBUTION OF COPIES.- Three copies, one of which shall be reproducible, of each report shall be forwarded to the procuring activity.

3.2 DISPOSITION OF QUALIFICATION TEST DATA.- Qualification test data shall be retained by the testing agency for 2 years, and furnished to the procuring activity or authorized representative upon request.

4. QUALITY ASSURANCE PROVISIONS

4.1 GENERAL.- All rocket motors and components submitted for qualification testing shall have been accepted using the acceptance method of the model specification. This acceptance shall demonstrate conformance with all requirements of the contract and specifications under which the rocket motors and components were built. Acceptance test data, as well as data on any rocket motors or components rejected during acceptance, shall be submitted as part of the qualification test data.

4.1.1 TEST METHOD AND APPARATUS.- Qualification test method and apparatus used shall be in accordance with that specified in the approved qualification test method document required by MIL-R-25532.

4.1.1.1 ACCURACY OF DATA.- For all rocket motor and component tests, the test apparatus shall be such as to insure that reported data will have a steady state accuracy of ± 2 percent of the maximum rated value, except that temperature shall be accurate to ± 2°F and weight shall be accurate to ± 0.2 percent. All apparatus shall be calibrated often enough to insure that this degree of accuracy is maintained. Calibration records shall be retained by the testing agency for two years, and furnished to the procuring activity upon request.

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4.1.1.2 AUTOMATIC RECORDING EQUIPMENT.- Automatic recording equipment of adequate response shall be used to obtain data during those parts of the rocket motor and component tests requiring the evaluation of time versus rocket motor or component variables.

4.1.2 MAINTENANCE.- Maintenance, adjustment, or replacement of components, or parts of components, shall not be permitted after initial acceptance.

4.2 GENERAL INSPECTION.- Rocket motors, components, and test apparatus shall be subject to inspection. At the option of the procuring activity, examinations may be made of random samples of rocket motors and components to determing whether they conform to all requirements of the contract and specifications under which they were built. These examinations may be conducted at any convenient time prior to or after testing. The results of all such examinations shall be submitted as part of the qualification test data.

4.3 WITNESSING.- All tests shall be subject to witnessing by an authorized representative(s) of the procuring activity.

4.4 ROCKET MOTOR TESTS

4.4.1 PREVIOUS TEST APPROVAL.- At the option of the procuring activity, individual tests specified herein may be waived based upon preliminary flight rating test firings of identical rocket motors. If such a waiver is granted a data summary sheet for each such firing shall be provided as a part of the qualification test report.

4.4.2 TEST CONDITIONS.- The following conditions shall apply for the rocket motor tests unless otherwise specified herein.

4.4.2.1 OPERATING CONDITIONS

4.4.2.1.1 TEMPERATURES.- Unless otherwise specified, all tests shall be conducted at room temperature as defined in 4.4.2.1.1.1.

4.4.2.1.1.1 ROOM TEMPERATURE.- Room temperature tests shall be conducted with the temperature of the rocket motor within the range of 10°F to 110°F.

4.4.2.1.1.2 LOW TEMPERATURE.- Low temperature tests shall be conducted with the temperature of the rocket motor at - 75°F or lower.

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4.4.2.1.1.3 NORMAL TEMPERATURE.- Normal temperature tests shall be conducted with the temperature of the rocket motor at + 59°F ± 5°F.

4.4.2.1.1.4 HIGH TEMPERATURE.- High temperature tests shall be conducted with the temperature of the rocket motor at + 170°F or higher.

4.4.2.1.2 PRESSURE.- Unless otherwise specified, all tests shall be conducted at ambient atmospheric pressure.

4.4.2.1.2.1 CHAMBER PRESSURE.- Chamber pressure measurements shall be made at the head end of the rocket motor.

4.4.2.1.3 TIMES

4.4.2.1.3.1 CONDITIONING TIME.- Conditioning time for the rocket motor shall be such that all parts of the rocket motor shall have reached a temperature within 5°F of the specified temperature. A rocket motor shall be considered conditioned when it has been continuously exposed to the specified temperature for the conditioning time, making suitable allowance for the starting temperature. During the conditioning time the temperature of the conditioning chamber shall not vary more than ± 5°F from the specified temperature.

4.4.2.1.3.2 TRANSFER TIME.- The rocket motor transfer time shall not exceed that specified in the model specification.

4.4.2.2 TEST GROUPS

4.4.2.2.1 ROCKET MOTORS.- Rocket motors shall be divided into test groups as shown on figure 3. Each propellant batch in the test quantity shall be represented as equally as possible in the test groups. Prior to the initiation of testing, a list showing the serial numbers of the rocket motors assigned to each test group shall be provided the procuring activity.

4.4.2.2.2 IGNITERS.- Igniters not permanently installed in the rocket motor shall be divided into test groups as shown on figure 3. Each igniter shall be identified in order to correspond to a specific rocket motor undergoing qualification test. Prior to the initiation of testing, a list showing the serial number of each igniter and the rocket motor to which it is assigned shall be provided the procuring activity. Igniters in all test groups except IV, V, and IX shall be packed in their individual shipping containers and shall be subjected to the same sequence of tests as the rocket motor with which they are identified. The igniters in groups IV and V shall be removed from individual containers,

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Test Group Number		I	II	III	IV	V	VI
Number of Rockets in Group		4	3	4	6	4	6
Paragraph No.	Title of Test						
4.4.3.1	Statistical Firing						
4.4.3.1.1	High Temperature	4					
4.4.3.1.2	Normal Temperature		3				
4.4.3.1.3	Low Temperature			4			
4.4.3.2	Sequential Treatment						
4.4.3.2.1	Temperature Cycling				6	4	6
4.4.3.2.2	Vibration						
	High Temperature				6		
	Normal Temperature					4	
	Low Temperature						6
4.4.3.2.3	Drop						
	High Temperature				6		
	Normal Temperature					4	
	Low Temperature						6
4.4.3.2.4	Altitude Cycling						
	High Temperature				6		
	Normal Temperature					4	
	Low Temperature						6
4.4.3.2.5	Humidity				3	2	3
4.4.3.2.6	Rain				3	2	3
4.4.3.2.7	Salt Spray				3	2	3
4.4.3.2.8	Sand and Dust				3	2	3
4.4.3.2.9	Static Firing						
	High Temperature				6		
	Normal Temperature					4	
	Low Temperature						6

FIGURE 3
GROUPS I THRU VI

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Test Group Number		VII	VIII	IX
Number of Rockets in Group		6	3	2
Paragraph No.	Title of Test			
4.4.3.3	Aging			
	Hot	6		
	Temperature Cycling	6		
	Vibration			
	High Temperature	3		
	Low Temperature	3		
4.4.3.3.1	Drop			
	High Temperature	3		
	Low Temperature	3		
	Static Firing			
	High Temperature	3		
	Low Temperature	3		
4.4.3.3.2	Cold		3	
4.4.3.4	Temperature Gradient Low Temperature			2

FIGURE 3 (continued)
GROUPS VII THRU IX

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Test Group Number		X	XI	XII	XIII	XIV	XV	XVI
Number of Rockets in Group		3	2	2	6	6	6	6
Paragraph No.	Title of Test							
4.4.3.5	Altitude Firing Normal Temperature Low Temperature	 1 2						
4.4.3.6	Acceleration Firing Stabilized Temperature Temperature Gradient		 1 1					
4.4.3.7	Safety Diaphragm Firing Burst High Temperature Low Temperature			 1 1				
4.4.3.8	Thrust Angle Control High Temperature Normal Temperature Low Temperature				 2 2 2			
4.4.3.9	Thrust Magnitude Control High Temperature Normal Temperature Low Temperature					 2 2 2		
4.4.3.10	Thrust Termination High Temperature Low Temperature						 3 3	
4.4.3.11	Thrust Reversal High Temperature Normal Temperature Low Temperature							 2 2 2

FIGURE 3 (continued)
GROUPS X THRU XVI

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installed in a simulated rocket motor, and subjected to the rain and humidity, or salt spray, and sand and dust tests, as applicable, immediately before firing. Unless otherwise specified in the model specification, the igniters in group IX shall be installed in the rocket motors of group IX during the temperature gradient test. The testing of the igniters will be scheduled so that completion of the test will coincide with the completion of the test on the corresponding rocket motor. Upon completion of the tests and after conditioning to the proper temperature, the igniter shall be removed from the shipping container or simulated rocket motor, installed in the corresponding rocket motor, and fired.

4.4.2.2.3 THRUST CONTROL DEVICES.- Rocket motors incorporating thrust control devices shall have such devices installed during all testing. These devices shall be operated in test groups XIII, XIV, XV, and XVI only.

4.4.2.3 DATA

4.4.2.3.1 PRELIMINARY DATA.- The following minimum preliminary data shall be recorded for each test firing.

a. Rocket motor designation

b. Rocket motor lot number

c. Rocket motor sublot number

d. Rocket motor serial number

e. Case serial number

f. Nozzle serial number

g. Propellant type or designation

h. Propellant lot number

i. Propellant batch number(s)

j. Propellant grain serial number

k. Igniter type

m. Igniter lot number

n. Igniter sublot number

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o. Squib resistance prior to conditioning or sequential treatment

p. Squib resistance after conditioning or sequential treatment

q. Motor weight before firing

r. Center of gravity location prior to firing

s. Nozzle expansion ratio

t. Nozzle throat area before firing

u. Conditioned temperature

v. Type of pretest treatment

w. Observed barometric pressure

x. Corrected barometric pressure

y. Time removed from conditioning

z. Ambient air temperature

4.4.2.3.2 TEST DATA.- The following minimum test data shall be recorded during each firing.

a. Thrust-time trace with point of firing switch actuation indicated

b. Chamber pressure-time trace with point of firing switch actuation indicated

c. Time fired

d. Igniter current

e. High speed thrust-time trace during ignition with point of firing switch actuation indicated

f. High speed chamber pressure-time trace during ignition with point of firing switch actuation indicated

4.4.2.3.3 POST FIRING DATA.- The following minimum data shall be taken as soon as possible after firing.

a. Rocket motor weight

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b. Center of gravity location

c. Nozzle throat area

4.4.2.3.4 REDUCED DATA.- The following data shall be calculated.

4.4.2.3.4.1 ROCKET MOTOR

a. Action time total impulse

b. Burn time total impulse

c. Deliverable total impulse

d. Specific impulse

e. Overall specific impulse

f. Maximum thrust

g. Maximum ignition thrust

h. Average thrust

i. Maximum chamber pressure

j. Maximum ignition pressure

k. Average chamber pressure

m. Action time

n. Burn time

o. Ignition delay time

p. Ignition rise time

q. Burn rate

4.4.2.3.4.2 THRUST ANGLE CONTROL DEVICE

a. Relation of actuating torque to control deflection

b. Relation of jet deflection to control deflection

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c. Relation of loss of action time total impulse to control deflection

d. Relation of normal force to control deflection

4.4.2.3.4.3 THRUST MAGNITUDE CONTROL DEVICE

a. Relation of thrust change to response time

b. Relation of actuating torque to thrust

4.4.2.3.4.4 THRUST TERMINATION DEVICE

a. Relation of thrust to time, after termination signal

4.4.2.3.4.5 THRUST REVERSAL DEVICE

a. Relation of thrust to time, after reversal signal

4.4.3 TESTS

4.4.3.1 STATISTICAL FIRING TESTS

4.4.3.1.1 HIGH TEMPERATURE.- The rocket motors of test group I shall be conditioned to and fired at the high temperature.

4.4.3.1.2 NORMAL TEMPERATURE.- The rocket motors of test group II shall be conditioned to and fired at the normal temperature.

4.4.3.1.3 LOW TEMPERATURE.- The rocket motors of test group III shall be conditioned to and fired at the low temperature.

4.4.3.2 SEQUENTIAL TREATMENT.- Test groups IV, V, and VI shall be subjected to the following sequential treatment. Treatment shall be in the order specified. Unless otherwise specified herein or by the procuring activity, the rocket motor shall be subjected to treatment in the operational configuration. If the igniter is removable, it shall be removed and the shipping closure installed.

4.4.3.2.1 TEMPERATURE CYCLING.- The rocket motor shall be conditioned successively to the high, low, high, low, high, and low temperatures. (The sequence may be reversed if desired.)

4.4.3.2.2 VIBRATION.- Vibration tests shall be conducted in accordance with procedure I unless the use of procedure II is approved by the procuring activity.

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4.4.3.2.2.1 PROCEDURE I.- The rocket motors of test groups IV, V, and VI shall be conditioned to and maintained at the high, normal, and low temperatures, respectively, during the test. The rocket motors shall be mounted to allow freedom of movement and excited through suitable nonresonant mountings. The vibration input shall be measured at these mounting points and shall be applied as follows:

a. Through the center of gravity in a direction parallel with the longitudinal axis of the rocket motor.

b. Through the center of gravity in a direction perpendicular to the longitudinal axis of the rocket motor.

4.4.3.2.2.1.1 SURVEY.- Excitation as noted in a and b above shall be applied to each of the rocket motors throughout the frequency range of 2 to 500 cps and critical resonances noted. Vibration measuring pickups shall be used at a sufficient number of locations on the rocket motor and components to detect and define critical resonances and associated vibration mode shapes. Vibration measurements shall be supplemented by visual observations with the aid of a stroboscope to assure proper definition of the vibration mode or by audible response.

4.4.3.2.2.1.2 ENDURANCE.- Each of the rocket motors shall be vibrated for 30 hours (15 hours at condition a and 15 hours at condition b) with an input of 0.100 inch double amplitude or ± 5g at each of the critical resonant frequencies determined in 4.4.3.2.2.1.1. If no critical resonances are detected, the rocket motor shall be vibrated for 15 hours with an input of 0.100 inch double amplitude or ± 5g at each of the following conditions:

Frequency, cps	Condition
20	a
20	b
200	a
200	b

If only one critical resonance is detected, the rocket motor shall be vibrated for 30 hours with an input of 0.100 inch double amplitude or ± 5g at that frequency under conditions a and b. The rocket motor shall also be vibrated at one of the above fixed frequencies, whichever is furthest from the critical resonance, under conditions a and b. In no case shall a rocket motor receive less than a total of 60 hours vibration.

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4.4.3.2.2.2 PROCEDURE II.- The rocket motors of test groups IV, V, and VI shall be conditioned to the high, normal, and low temperatures, respectively, and placed in the shipping containers. The shipping container shall be placed on a truck and transported over 500 miles of secondary roads. The type of truck and the route selected shall be subject to approval by the procuring activity. The test shall be conducted at the prevailing ambient temperature conditions.

4.4.3.2.3 DROP.- Procedure I shall be used for all rocket motors weighing 200 pounds or less and all igniters. Procedure II shall be used for all rocket motors weighing more than 200 pounds.

4.4.3.2.3.1 PROCEDURE I.- The rocket motors of test groups IV, V, and VI shall be conditioned to the high, normal, and low temperatures, respectively, prior to each drop. Conditioned rocket motors shall be subjected to a drop of 4 feet onto solid reinforced concrete as follows:

a. Two (one in group V) rocket motors with longitudinal axis vertical - impact on head end.

b. Two (one in group V) rocket motors with longitudinal axis vertical - impact on nozzle end.

c. Two rocket motors with longitudinal axis horizontal - impact on side.

4.4.3.2.3.2 PROCEDURE II.- The rocket motors of test groups IV, V, and VI shall be conditioned to the high, normal, and low temperatures, respectively, and placed in the shipping container. The rocket motors shall then be subjected to the edgewise drop tests, cornerwise drop tests, and either the pendulum-impact tests or the incline-impact tests specified in MIL-P-7936. Level A shall be used unless otherwise specified by the procuring activity.

4.4.3.2.4 ALTITUDE CYCLING.- The rocket motors shall be assembled as for packaging with all shipping closures in place when subjected to this cycling. The rocket motors of test groups IV, V, and VI shall be conditioned to and maintained at the high, normal, and low temperatures, respectively, during the cycling. The external pressure shall be maintained at ambient for 30 minutes, reduced to 2.1 ± 0.2 inches of mercury and maintained for 30 minutes, then raised to ambient. The pressure change shall be completed in less than 5 minutes, and the elapsed time shall be credited against the next 30 minute period. This cycle shall be repeated five times.

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4.4.3.2.5 HUMIDITY.- One-half of the rocket motors from test groups IV, V, and VI shall be subjected to a humidity test in accordance with procedure III of MIL-E-5272, except that the rocket motor and components shall not be operated at the completion of the test.

4.4.3.2.6 RAIN.- The rocket motors that have been subjected to humidity shall be subjected to a rain test in accordance with procedure I of MIL-E-5272.

4.4.3.2.7 SALT SPRAY.- One-half of the rocket motors from test groups IV, V, and VI (not rocket motors that have been subjected to humidity or rain) shall be subjected to a salt spray test in accordance with procedure I of MIL-E-5272, except that the rocket motor and components shall not be operated at the completion of the test.

4.4.3.2.8 SAND AND DUST.- The rocket motors that have been subjected to salt spray shall be subjected to a sand and dust test in accordance with procedure II of MIL-E-5272, except that the rocket motor and components shall not be operated at the completion of the test.

4.4.3.2.9 STATIC FIRING.- The rocket motors of test groups IV, V, and VI shall be conditioned to the high, normal, and low temperatures, respectively. Conditioning shall be initiated within 2 hours after sequential treatment is completed. After conditioning, the rocket motors shall be static fired.

4.4.3.3 AGING TESTS

4.4.3.3.1 HOT.- The rocket motors of test group VII shall be stored for 45 days at the high temperature. The rocket motors shall then be divided into two subgroups of three each (designated subgroups A and B) and subjected to the following tests in the order specified.

a. Temperature cycling.- Use procedure specified in 4.4.3.2.1 for the rocket motors of both subgroup A and subgroup B.

b. Vibration.- Use procedure specified in 4.4.3.2.2. The rocket motors of subgroups A and B shall be conditioned to and maintained at the high and low temperatures, respectively, during the test.

c. Drop.- Use procedure specified in 4.4.3.2.3. The rocket motors of subgroups A and B shall be conditioned to and maintained at the high and low temperatures, respectively, during the test.

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d. Static firing.- The rocket motors of subgroups A and B shall be conditioned to the high and low temperatures, respectively, and static fired.

4.4.3.3.2 COLD.- The rocket motors of test group VIII shall be conditioned to 0°F. Each rocket motor shall then be stored for seven days at each of the following temperatures, in the order specified: 0°F, - 15°F, - 30°F, - 45°F, - 60°F, and - 75°F. The rocket motor temperature shall not rise at any time during the test. After completion of this treatment, the rocket motors shall be removed from conditioning and static fired.

4.4.3.4 TEMPERATURE GRADIENT TEST.- The rocket motors of test group IX shall be conditioned to the low temperature, then exposed to the high temperature for the time specified in the model specification, which will cause the maximum temperature gradient to exist across the propellant grain. The rocket motors shall then be static fired. A nozzle closure shall be installed during conditioning, regardless of whether one is normally used.

4.4.3.5 ALTITUDE FIRING TEST.- One rocket motor of test group X shall be conditioned to the normal temperature, and two rocket motors of test group X shall be conditioned to the low temperature. These rocket motors shall be static fired at a simulated altitude equal to the maximum ignition altitude specified in the model specification.

4.4.3.6 ACCELERATION FIRING TEST.- The rocket motors of test group XI shall be fitted with stabilizing fins and adequate ballast to produce a maximum acceleration in the longitudinal direction under free flight conditions of from 1.20 to 1.30 times the maximum longitudinal acceleration specified in the model specification. One rocket motor shall be conditioned to the high temperature and fired under free flight conditions. One rocket motor shall be conditioned to the low temperature, then exposed to the high temperature for the time specified in the model specification which will cause the maximum temperature gradient to exist across the propellant grain, and then fired under free flight conditions. As alternate test procedures, firings on a ballistic centrifuge or sled will be acceptable providing the temperature and acceleration requirements specified herein are met.

4.4.3.7 SAFETY DIAPHRAGM FIRING BURST TEST.- The rocket motors of test group XII shall be conditioned one each to the high and low temperature. The rocket motors shall then be subjected to a firing test in which the exhaust flow through the nozzle is interrupted to cause the safety diaphragm to rupture

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during the final 30 percent of the burn time. The interruption of the exhaust flow shall be accomplished by an externally applied pintle or other equivalent means, and the rate of pressure rise within the rocket motor shall not exceed 20,000 psi per second.

4.4.3.8 THRUST ANGLE CONTROL DEVICE.- The thrust angle control devices from groups A and B which have been subjected to the tests specified in 4.5.3.1 shall be installed on the rocket motors of test group XIII. Two rocket motors each shall be conditioned to the high, normal, and low temperatures and static fired. The thrust angle control device operating cycle shall be as specified by the procuring activity.

4.4.3.9 THRUST MAGNITUDE CONTROL DEVICE.- The thrust magnitude control devices from groups A and B which have been subjected to the tests specified in 4.5.3.1 shall be installed on the rocket motors of test group XIV. Two rocket motors each shall be conditioned to the high, normal, and low temperatures and static fired. The thrust magnitude control device operating cycle shall be as specified by the procuring activity.

4.4.3.10 THRUST TERMINATION DEVICE.- Three rocket motors each from test group XV shall be conditioned to the high and the low temperatures and static fired. The thrust termination device shall be actuated to cause thrust termination during what would normally constitute the final 30 percent of burn time.

4.4.3.11 THRUST REVERSAL DEVICE.- Two rocket motors each from test group XVI shall be conditioned to the high, normal, and low temperatures and static fired. The thrust reversal device shall be actuated to cause thrust reversal during what would normally constitute the final 30 percent of burn time.

4.4.3.12 ADDITIONAL TESTS.- Additional rocket motor tests for the purpose of qualifying other special features of the rocket motor and components may be required by the procuring activity. These tests shall be as specified in the model specification.

4.5 ROCKET MOTOR COMPONENT TESTS.- Approval of the component tests shall be required prior to the approval of the rocket motor as a service type or model.

4.5.1 PREVIOUS COMPONENT TEST APPROVAL.- Rocket motor components requiring testing as specified herein may have these tests waived at the option of the procuring activity, if the component has been previously subjected to this testing and qualified. Such components must be substantially identical to the respective components previously qualified with the exception of mounting provisions.

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If such a waiver is granted, information on the components for which previous approval was obtained shall be provided as a part of the qualification test report.

4.5.2 TEST CONDITIONS.- The following conditions shall apply for the rocket motor component tests unless otherwise specified herein.

4.5.2.1 OPERATING CONDITIONS

4.5.2.1.1 GENERAL.- Prior to the start of testing all components shall be cleaned of oil, grease, or other corrosion preventive compound used during shipment. Insofar as practicable, components shall be mounted in their normal positions as mounted on the rocket motor. Components shall be subjected to operating loads simulating those encountered on the rocket motor. Sufficient instrumentation shall be provided to indicate the performance of each component and to indicate that the functional relationships of components are maintained as required by the applicable test schedule. Functional checks shall be performed at the end of each test to indicate that no calibrated component has changed its calibration and that the function of uncalibrated components is unimpaired. All components shall be supplied with such fluids as they normally handle or contact. There shall be no traces of external fluid leakage from any component.

4.5.2.1.2 TEMPERATURES.- Unless otherwise specified, all tests shall be conducted at room temperature as defined in 4.5.2.1.2.1.

4.5.2.1.2.1 ROOM TEMPERATURE.- Room temperature tests shall be conducted with the temperature of the component within the range of 10°F to 110°F.

4.5.2.1.2.2 LOW TEMPERATURE.- Low temperature tests shall be conducted with the temperature of the component at - 75°F or lower.

4.5.2.1.2.3 NORMAL TEMPERATURE.- Normal temperature tests shall be conducted with the temperature of the component at 59°F ± 5°F.

4.5.2.1.2.4 HIGH TEMPERATURE.- High temperature tests shall be conducted with the temperature of the component at 170°F or higher.

4.5.2.1.3 PRESSURE.- Unless otherwise specified, all tests shall be conducted at ambient atmospheric pressure.

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4.5.2.1.4 TIMES

4.5.2.1.4.1 CONDITIONING TIME.- Conditioning time for the component shall be such that all parts of the component shall have reached a temperature within 5°F of the specified temperature. A component shall be considered conditioned when it has been continuously exposed to the specified temperature for the conditioning time, making suitable allowance for the starting temperature. During the conditioning time the temperature of the conditioning chamber shall not vary more than ± 5°F from the specified temperature.

4.5.2.1.4.2 TRANSFER TIME.- The component transfer time shall not exceed 5 minutes.

4.5.2.2 DATA.- The following minimum data shall be recorded for each component test.

a. Component model designation or nomenclature

b. Component serial number

c. Component sublot number

d. Conditioned temperature

e. Type of test

f. Ambient air temperature

g. Barometric pressure

h. Calibration data (if applicable)

i. Recalibration data (if applicable)

4.5.3 TESTS.- Prior to the initiation of the component tests, each component for which the establishment of input-output relationships is a function of the component shall be subjected to a calibration. The calibration shall be extensive enough to cover the entire steady state and dynamic ranges of operation of the component on the rocket motor and shall indicate conformance with the design tolerance range of the component. Components not subject to calibration shall be operated under normal operating conditions to demonstrate satisfactory functioning. Each calibration shall be recorded.

4.5.3.1 THRUST CONTROL DEVICES.- Nine devices of each type shall be divided into three test groups of three. Group A shall be subjected to accelerated aging, humidity, and rain. Group B

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shall be subjected to accelerated aging, salt spray, and sand and dust. Group C shall be subjected to fungus. Treatment shall be in the order specified.

4.5.3.1.1 ACCELERATED AGING.- Upon completion of the component calibrations, all components shall be placed, completely assembled and filled with working fluid (if applicable), in an air oven and maintained in an ambient temperature of not less than 160°F for a period of time not less than 168 hours.

4.5.3.1.2 HUMIDITY.- Components shall be subjected to a humidity test in accordance with procedure III of MIL-E-5272.

4.5.3.1.3 RAIN.- Components shall be subjected to a rain test in accordance with procedure I of MIL-E-5272.

4.5.3.1.4 SALT SPRAY.- Components shall be subjected to a salt spray test in accordance with procedure I of MIL-E-5272.

4.5.3.1.5 SAND AND DUST.- Components shall be subjected to a sand and dust test in accordance with procedure II of MIL-E-5272.

4.5.3.1.6 FUNGUS.- Components shall be subjected to a fungus test in accordance with procedure I of MIL-E-5272.

4.5.3.2 POWER SYSTEMS

4.5.3.2.1 ELECTRICAL SYSTEM

4.5.3.2.1.1 COMPONENTS.- Electrical system components shall be subjected to the qualification tests specified in the applicable military specification for the component.

4.5.3.2.2 HYDRAULIC SYSTEM.- The complete hydraulic system shall be subjected to the tests specified in MIL-H-25475.

4.5.3.2.2.1 COMPONENTS.- Hydraulic system components shall be subjected to the qualification tests specified in MIL-H-8775 and the applicable military specification for the component.

4.5.3.2.3 PNEUMATIC SYSTEM

4.5.3.2.3.1 COMPONENTS.- Pneumatic system components shall be subjected to the qualification tests specified in MIL-P-8564 and the applicable military specification for the component.

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4.5.3.3 OTHER ROCKET MOTOR COMPONENTS

4.5.3.3.1 CASES.- Five complete cases shall be subjected to a hydrostatic burst test. Safety diaphragms shall be plugged if applicable. Pressures causing initial yield and failure shall be determined. The cases shall provide the maximum representation of case lots used in this qualification test.

4.5.3.3.2 SAFETY DIAPHRAGM ASSEMBLY.- Forty-five safety diaphragm assemblies, fifteen each conditioned to the high, normal, and low temperatures, shall be subjected to a hydrostatic burst test. The pressure at which failures occur shall be determined. The safety diaphragms shall provide the maximum representation of safety diaphragm lots used in this qualification test.

4.5.3.3.3 ATTACHMENT FITTINGS

4.5.3.3.3.1 ROCKET MOTOR.- Five complete case assemblies shall be subjected to an attachment fitting test. The cases shall be mounted to simulate vehicle installation and the static load applied in the same direction as the maximum thrust attainable. The loads causing initial yield and failure of the attachment fittings shall be determined.

4.5.3.3.3.2 THRUST CONTROL DEVICES.- Five sets of each type of thrust control device shall be subjected to an attachment fitting test. The devices shall be mounted to simulate normal installation and the static load(s) applied in the same direction(s) as loads are applied during rocket motor operation. The loads causing initial yield and failure of the attachment fittings shall be determined.

4.5.3.3.4 SQUIBS AND EXPLOSIVE COMPONENTS.- The autoignition temperature of all squibs and explosive components shall be determined as follows: The squib or explosive component shall be placed in an oven that has been preheated to the test temperature. The highest test temperature at which the component will not autoignite after 30 minutes is reported as the autoignition temperature, 30-minute test. The highest test temperature at which the sample will not autoignite after 8 hours is reported as the autoignition temperature, 8-hour test. Six samples of each shall be tested.

4.5.3.3.5 ADDITIONAL TESTS.- Additional component tests for the purpose of qualifying other special features may be required by the procuring activity. These tests shall be as specified in the model specification.

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4.6 TEARDOWN INSPECTION.- After completion of the required tests the rocket motors and components shall be disassembled for measurements, examination of parts, and photographs.

4.7 TEST COMPLETION.- The qualification test is considered complete when the rocket motor and components have been subjected to the required tests and inspections and the required reports have been submitted to the procuring activity for approval.

5. NOTES

5.1 INTENDED USE.- The tests contained in this specification are intended to cover the qualification testing of rocket motors.

5.2 SYMBOLS AND DEFINITIONS.- The symbols and definitions used in this specification are as specified in MIL-R-25532.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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